

CBSE Test Paper 01
Chapter 4 Motion in A Plane

1. The speed of a projectile when it is at its greatest height is $\sqrt{\frac{2}{5}}$ times its speed at half the max height. What is the angle of projection? **1**
 - a. 60°
 - b. 90°
 - c. 15°
 - d. 45°
 2. if particles A and B are moving with velocities $\mathbf{v_A}$ and $\mathbf{v_B}$ (each with respect to some common frame of reference, say ground.). Then, velocity of particle A relative to that of B is: **1**
 - a. $\mathbf{v_{AB}} = \mathbf{v_A} + \mathbf{v_B}$
 - b. $\mathbf{v_{AB}} = \mathbf{v_A} - \mathbf{v_B}$
 - c. $\mathbf{v_{AB}} = -\mathbf{v_A} + \mathbf{v_B}$
 - d. $\mathbf{v_{AB}} = -\mathbf{v_A} - \mathbf{v_B}$
 3. An airplane's compass indicates that it is headed due north, and its airspeed indicator shows that it is moving through the air at 240 km/h. If there is a 100-km h wind from west to east. In what direction should the pilot head to travel due north? What will be her velocity relative to the earth? **1**
 - a. 35° W of N, 198 km/h
 - b. 28° W of N, 210 km/h
 - c. 25° W of N, 218 km/h
 - d. 29° W of N, 200 km/h
 4. The path of a projectile is **1**
 - a. cubic
 - b. straight line
 - c. hyperbola
 - d. a parabola
 5. Which of the following algebraic operations on vectors not permissible? **1**
 - a. adding a scalar component of the vector to the same vector
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- b. multiplying any vector by any scalar,
c. adding any two vectors
d. multiplying the sum of vectors A and B by a scalar
6. What will be the net effect on maximum height of a projectile when its angle of projection is changed from 30° to 60° , keeping the same initial velocity of projection?
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7. If $|\mathbf{A} \times \mathbf{B}| = \mathbf{A} \cdot \mathbf{B}$, what is the angle between A and B? **1**
8. Three vectors not lying in a plane can never end up to give a null vector. Justify. **1**
9. A man can swim with a speed of 4 km/h in still water. How long does he take to cross a river 1 km wide, if the river flows steadily, 3 km/h and he makes his strokes normal to the river current? How far down the river does he go when he reaches the other bank? **2**
10. On a certain day, rain was falling vertically with a speed of 35 m/s. A wind started blowing after sometime with a speed of 12 m/s in East to West direction. In which direction should a boy waiting at a bus stop hold his umbrella? **2**
11. What is the angle of projection at which horizontal range and maximum height are equal? **2**
12. A boy stands at 78.4 m from a building and throws a ball which just enters a window 39.2 m above the ground. Calculate the velocity of projection of the ball. **3**
13. Find the angle between the vectors $\mathbf{A} = \hat{\mathbf{i}} - 2\hat{\mathbf{j}} - \hat{\mathbf{k}}$ and $\mathbf{B} = -\hat{\mathbf{i}} + \hat{\mathbf{j}} - 2\hat{\mathbf{k}}$. **3**
14. An aircraft is flying at a height of 3400 m above the ground. If the angle subtended at a ground observation point by the aircraft positions 10.0 s apart is 30° , what is the speed of the aircraft? **3**
15. On an open ground, a motorist follows a track that turns to his left by an angle of 60° after every 500 m. Starting from a given turn, specify the displacement of the motorist at the third, sixth and eighth turn. Compare the magnitude of the displacement with the total path length covered by the motorist in each case. **5**
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